Design and Fabrication of Langmuir Probe Circuit for Measurement of Plasma Edge Electron Temperature and Density in IR-T1 Tokamak

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Abstract

To achieve the edge plasma properties, 2 electronic circuit have been fabricated and a set of 4 single Langmuir probes installed in Iran –Tokamak1 (IR-T1). The Floating Potential has been measured without applying any voltage, Ion Saturation Current by a negative DC power supply and then by applying a sweep voltage to tips, I-V curve and so electron temperature is obtained. Then we compare these results to investigate of some properties of edge plasma.

Keywords: Langmuir Probe, Floating Potential, Electron Temperature, Ion Density

1) Introduction

Langmuir probe is a fundamental edge diagnostic which is used to determine several basic properties of plasma edge such as electron temperature(T_e), electron density (n_e), and floating potential(V_f).

Electric probes (Langmuir probes) are relatively simple devices, but the theory underlying it is complicated. Basically, the electric probe consists of one or more small metallic

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electrodes inserted in to the plasma. In a single probe configuration (which is used in this experiment), a single electrode is inserted into the torous and these experiments is done.

1. without any voltage and just wih a proper electronic circuit, Floating potential is obtained.

2. a DC power supply with various negative potential biased the tips and the Ion saturation Current is got.

3. a sweep voltage is applied to tips and current collected by the probe is plotted as a function of potential. This plot which is called I-V characteristic of probe, is used to calculate electron temperature.

2) Theory of Langmuir probe

By applying a sweep voltage to the Langmuir probe which is referred to a grounded electrode (like the conducting wall chamber), we can measure the current in the probe and plot I-V curve. A typical I-V curve from single Langmuir probe is shown in figure(1)(1).

The qualitative behavior of the I-V curve can be explained as follow. For large negative and positive values of the probe voltage, saturation currents called ion saturation current Iis and electron saturation current Ies respectively. The point at which the curve crosses the V axis is called floating potential Vf, where the collected ion and electron, exactly balances each other and the net current is zero. At the space potential (Vp) the net current takes a sharp turn, called knee, and then current saturates at the electron saturation current (Ies).

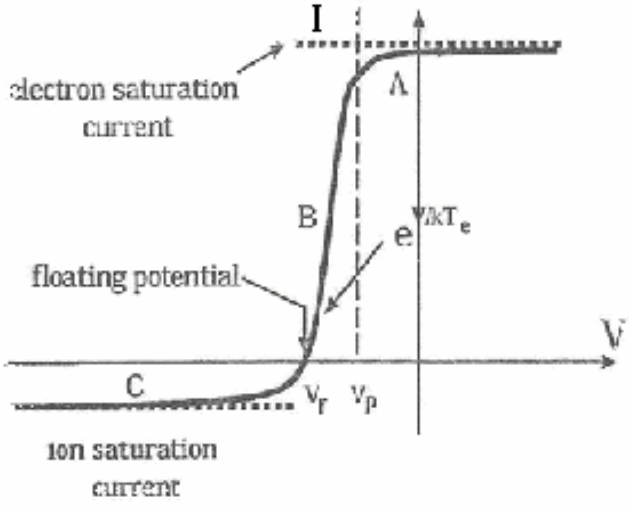


Fig.1. A typical I-V Curve For Single probe

As figure (1) shows, with increasing voltage above the Ion saturation region, the probe current increases exponentially. This region is called transition region. Thus in the region between Vf and Vp, the probe potential is still negative, relative to the plasma potential (Vp), however the probe attracts increasingly more electrons, so that the net current to the probe is electron current. If the electron distribution was Maxwellian (as it often is), the shape of the curve, after Ion contribution is subtracted, would be exponential. Now according to equation (1) and I-V curve, and using a nonlinear least-square fitting, we can determine Te (2,3).

$I=Iis\left(1-e^{e(V-V_f)/KT_e}\right)$ (1)

Also Ion density is obtained from:

Iis=ni eA(2KTe/mi)1/2 (2)

(Where A is the exposed area of the probe tip.)

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3) Probe Structure

Since the probe is immersed in a harsh environment, we used special techniques to protect it from the plasma. The probe that had been installed in IR-T1 tokamak is made of a set of 4 single probe (Fig. 2).

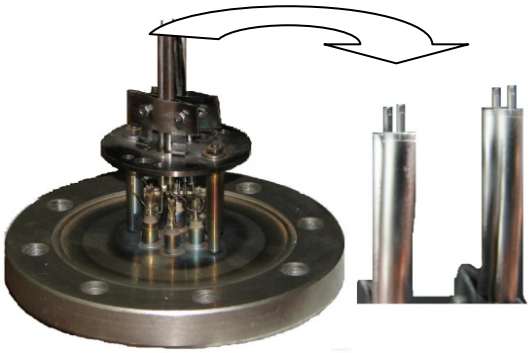


Fig.2. Langmuir Probe and base, pins 1&2 are shorter

The tips have been made from tungsten rod that can bear high-temperature. In order to protect tungsten from the plasma, except for a short length of exposed tip, the rod is threaded into a thin ceramic tube with 3 mm diameter. In addition to protect pin, this insulator prevents from electrical contact between the tips and conducting wall of torus. Diameter of tip is 0.6 mm with 3.9mm length. Distance between 2 same height tips is 3.9mm and distance between short and high tips is 8.4 mm. For reaching to a better thermal conducting between tip and torus, a metallic shield covers isolator (of course because the plasma is pulsed, the probe can withstand the heat.). The whole of probe assembly is encased in torus under vacuum.

4) Electronic Circuit For Single Probe

Two power supplies for operating in different modes have been designed. With the first circuit we can measure both floating potential and Ion saturation current (Fig.3). Other system is a sweep voltage power supply. Operational Frequency is 0-1000 Hz and the voltage range is -120 to +120 volts. By using this power supply we can get I-V curve of tips.

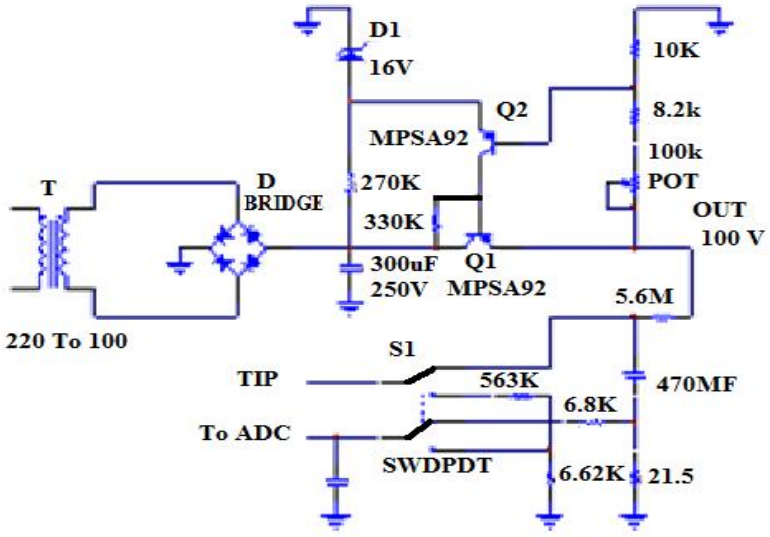


Fig.(3). Electronic Circuit For floating potential & Ion Saturation Current

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5) IR-T1 Tokamak

IR-T1 Tokamak is an air core tokamak without copper shell. Table 1 gives the parameters of the IR-T1 tokamak.

TABLE 1: Different parameters of IR-T1 Tokamak

Parameters	Value
Major Radius	45cm
Minor Radius	12.5cm
Plasma Radius	11.5cm
Toroidal Field	<1T
Plasma Current	<40kA
$t_{discharge}$	<40ms
$n_e \sim n_i$	$(0.7\text{-}3.0)\times10^{13}cm^{-3}$

6) Experiments and Results

At the beginning, in order to obtain floating potential for 4 pins, the system was sat up with using the first power supply. In the first mode (floating potential) without applying any voltage to pins, current is measured. Figure (4) shows results of this measurement for 4 pins.

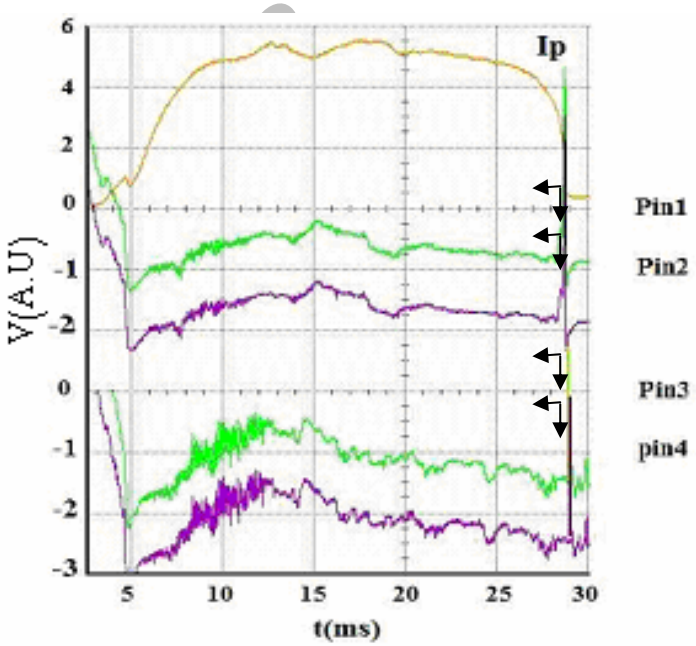


Fig.(4). Floating potential For 4 Pins and I_p

In this shot we get approximately 30.1Vfor floating potential for pins 1&2 and 51.6 V for Pins 3&4. Next, in the other mode (Ion Saturation Current) and with different negative voltages we reach to Ion Saturation current in less than -60V. (Fig.5)

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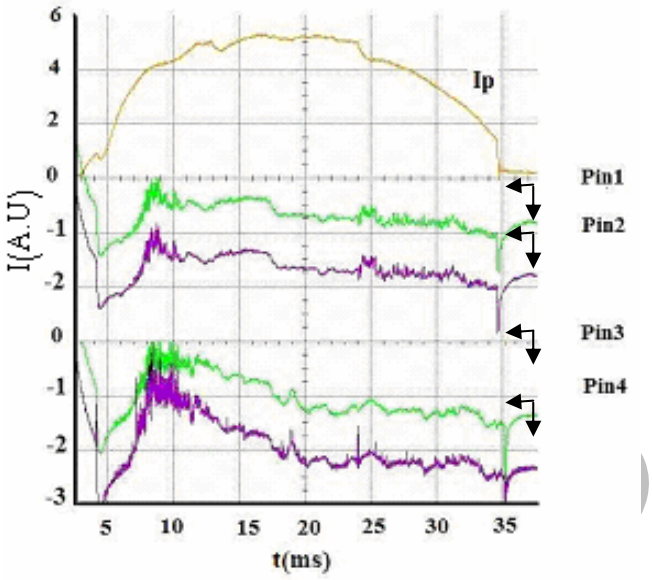


Fig.(5). I.S.C for 4 pins and I_p

By using this diagram and circuit of I.S.C we obtain about 16.6 mA for ion saturation current for Pins 1&2 and 28.6mA for Pins 3&4.Then a sweep voltage applied to the probe. (Fig.6)

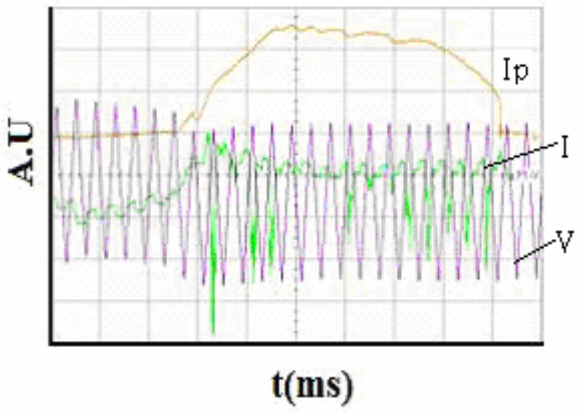
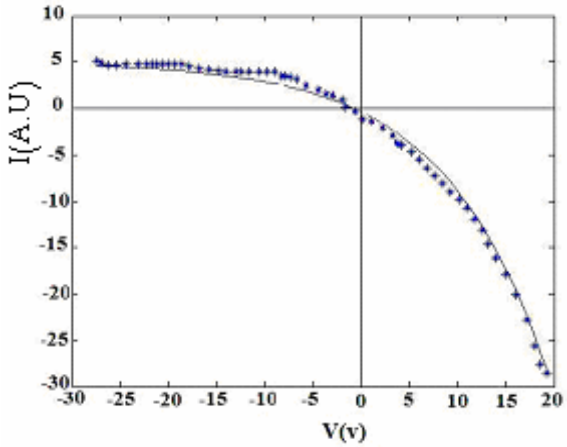


Fig.(6) I_P, Sweep Voltage & Probe Current for Pin 1 in a typical Shot

Figure (7) shows I-V curve for pin 1. With a nonlinear least-squire fitting in MATLAB program, electron temperature was obtained 10.5eV. According to eq.(2) and Te that have been calculated from I-V curve, we can obtain Ion density in quasi neutrality condition.

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Fig(7). I-V Curve for Pin 4 in a typical Shot

If we consider A=1.74mm2, and for pin4 with Te=10.5eV, then we obtain ni=2.3×1012cm-3, and for pin2 with Te=10.43eV we obtain ni=1.83×1012 cm-3 .

7) Conclusion

Regarding to the first results without position control of plasma column, we can compare properties of edge plasma such as floating potential, electron density and temperature in 4 points near to limiter. As the results show, the floating potential, Ion density and electron temperature for deeper pins are more than closers.

8) References

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